CheckPointOne Report

October 21, 2026

Section 1: Meta Information

1. **Open Project Title:** Investigating the Relationship Between Students' Lifestyle Choices and Academic Performance
2. **Group Number:** "33"
3. **Mentor Name:** Avadh Hindocha
4. **Member Names and Roll Numbers:** Yaswanth (25B0993), Rutvik Sai (25B1016), Goutham Pinnani (25B1019), Chakri (25B1070)
5. **GitHub Repository Link:** https://github.com/svschakri/CS-215-DAI_project

Section 2: Progress So Far

1. **Descriptive Statistics:** The dataset is the publicly downloadable Kaggle Student Lifestyle Dataset (no special access permissions needed). Each row is one student and the columns are `Student_ID`, five daily-hours variables (study, extracurricular, sleep, social, physical activity), `GPA`, and the categorical `Stress_Level` (Low/Moderate/High). These five daily-hours variables nicely sumup to 24hrs for every student. We have computed the size of the data, the mean, standard deviation, minimum and maximum of each numeric variable, and the number of students in each stress group. We also plotted the distribution of each variable, compared every variable and GPA across stress levels with box plots, and computed group-wise averages and the correlation of each lifestyle variable with GPA.
2. **Data Cleaning Steps Completed:** `Student_ID` is unique, so no rows were dropped or imputed. Verified that all values lie in plausible ranges and that `Stress_Level` has only the three expected categories. `Student_ID` is excluded from all analyses. We also found that the five hour variables sum to exactly 24 for every student, so they are not independent (see Section 2, item 4).
3. **Division and Revision of Responsibilities: Yaswanth**: Data collection, verifying the data, checkpointOne report and the CLT simulation; **Rutvik Sai**: box plots of each variable against stress level; **Goutham Pinnani**: code readability and proposing hypotheses for later testing; **Chakri**: histograms of each variable and GPA. We plan to do hypothesis testing and regression next, and will share this work across the group.
4. **Fine-Grained Problem Statement Refinement:**
   - *Goal 1:* Analysying the corellations of multiple daily activities with one another.
   - *Goal 2:* given a target CGPA (e.g., 3.0), recommend a daily 24-hour time allocation that meets the target while maximizing the time left for non-academic activities (social, extracurricular, physical activity), with sleep treated as a minimum requirement and `Stress_Level` as an indicator of well-being.

   We intend to do some hypothesis testing, but have not yet decided which hypotheses to test.
5. **Updated Tracking Spreadsheet:** Tracking sheet.
   - Meta-information row: added (title, group number, mentor, members, repository link).
   - Mentor edit access: sheet shared with the mentor with EDIT permission.
6. **Blockers/Challenges:** We would like guidance on (i) which hypotheses and statistical tests are appropriate given the observed distributions, (ii) handling the fixed 24-hour total which makes the parameters not independent and (iii) choosing confidence levels and intervals.